# Climate-Integrated Street Cleaning Routing Problem (MILP Formulation)

## 1 Notation

Let $\mathcal{N} = \{0, \ldots, N-1\}$ be the set of junctions and $\mathcal{J} = \{0, \ldots, M-1\}$ the set of streets in the city network. The streets are partitioned into three categories: $\mathcal{J}_M$ (mandatory), $\mathcal{J}_O$ (optional), and $\mathcal{J}_C$ (connectors), so that $\mathcal{J} = \mathcal{J}_M \cup \mathcal{J}_O \cup \mathcal{J}_C$. Let $\mathcal{A}$ be the set of directed arcs obtained from the street network, where each two-way street generates two arcs and each one-way street generates one. For a street $j \in \mathcal{J}$, denote by $A_j \subseteq \mathcal{A}$ the set of arcs corresponding to $j$. For a junction $u \in \mathcal{N}$, let $\delta^+(u)$ and $\delta^-(u)$ be the sets of arcs leaving and entering $u$, respectively.

Let $\mathcal{K} = \{0, \ldots, C-1\}$ be the fleet of cleaning vehicles. For each arc $a \in \mathcal{A}$, denote by $c_a$ its traversal time in seconds. For each street $j \in \mathcal{J}$, denote by $\ell_j$ its length in metres and by $r_j$ its cleaning requirement in litres per kilometre. For each vehicle $k \in \mathcal{K}$, denote by $q_k$ its water capacity in litres per kilometre. A vehicle $k$ can clean street $j$ only if $q_k \geq r_j$; connector streets ($j \in \mathcal{J}_C$) cannot be cleaned.

Let $T$ be the shared time limit for every vehicle, $S \in \mathcal{N}$ the depot junction at which all vehicles start and must return, and $W \geq 0$ the water-waste penalty factor. Define $U_a = \min(\lfloor T/c_a \rfloor, 50)$ as the traversal upper bound for arc $a$ and $\mathcal{M} = 1 + \sum_{a \in \mathcal{A}} U_a$ as the Big-M constant.

Denote by $f_{a,k}$, $a \in \mathcal{A}$, $k \in \mathcal{K}$, the integer decision variable representing the number of times vehicle $k$ traverses arc $a$. Denote by $x_{j,k}$, $j \in \mathcal{J} \setminus \mathcal{J}_C$, $k \in \mathcal{K}$, the binary decision variable which takes the value 1 if vehicle $k$ cleans street $j$, and 0 otherwise. If $q_k < r_j$, the variable $x_{j,k}$ is fixed to 0. Finally, denote by $y_j$, $j \in \mathcal{J} \setminus \mathcal{J}_C$, the binary decision variable which takes the value 1 if street $j$ is cleaned by any vehicle, and 0 otherwise.

| Symbol | Description |
|---|---|
| $\mathcal{N} = \{0, \ldots, N-1\}$ | Junctions (nodes) |
| $\mathcal{J} = \{0, \ldots, M-1\}$ | Streets (edges) |
| $\mathcal{J}_M \subset \mathcal{J}$ | Mandatory streets |
| $\mathcal{J}_O \subset \mathcal{J}$ | Optional streets |
| $\mathcal{J}_C \subset \mathcal{J}$ | Connector streets ($\mathcal{J} = \mathcal{J}_M \cup \mathcal{J}_O \cup \mathcal{J}_C$) |
| $\mathcal{A}$ | Directed arcs (a two-way street generates two arcs) |
| $A_j \subseteq \mathcal{A}$ | Set of arcs corresponding to street $j$ |
| $\mathcal{K} = \{0, \ldots, C-1\}$ | Vehicles |
| $\delta^+(u), \delta^-(u)$ | Outgoing / incoming arcs at node $u$ |

| Symbol | Description |
|---|---|
| $T$ | Shared time limit for every vehicle (seconds) |
| $S$ | Depot junction |
| $W$ | Water-waste penalty factor |
| $c_a$ | Traversal time of arc $a$ (seconds) |
| $\ell_j$ | Length of street $j$ (metres) |
| $r_j$ | Cleaning requirement of street $j$ (litres / km) |
| $q_k$ | Water capacity of vehicle $k$ (litres / km) |
| $U_a$ | Traversal upper bound for arc $a$: $U_a = \min\big(\lfloor T/c_a \rfloor,\ 50\big)$ |
| $\mathcal{M}$ | Big-M constant: $\mathcal{M} = 1 + \sum_{a \in \mathcal{A}} U_a$ |

## 2 Decision variables

$$f_{a,k} \in \mathbb{Z}_{\geq 0} \qquad \forall\, a \in \mathcal{A},\ k \in \mathcal{K} \qquad \text{(flow variable)}$$

$$x_{j,k} \in \{0,1\} \qquad \forall\, j \in \mathcal{J} \setminus \mathcal{J}_C,\ k \in \mathcal{K} \qquad \text{(cleaning assignment)}$$

$$y_j \in \{0,1\} \qquad \forall\, j \in \mathcal{J} \setminus \mathcal{J}_C \qquad \text{(cleaned indicator)}$$

- $f_{a,k}$ counts how many times vehicle $k$ traverses arc $a$.
- $x_{j,k} = 1$ if vehicle $k$ cleans street $j$.
- $y_j = 1$ if street $j$ is cleaned by *any* vehicle.

**Note.** If vehicle $k$ cannot clean street $j$ (i.e. $q_k < r_j$, or $j \in \mathcal{J}_C$), $x_{j,k}$ is fixed to 0 and not created as a decision variable.

## 3 Formulation

Using the above definitions, the Climate-Integrated Street Cleaning Routing Problem can be formulated as follows:

$$\text{Maximise} \quad \sum_{j \in \mathcal{J} \setminus \mathcal{J}_C} \ell_j \, y_j \; - \; W \sum_{j \in \mathcal{J} \setminus \mathcal{J}_C} \sum_{k \in \mathcal{K}} \frac{(q_k - r_j)\, \ell_j}{1000} \, x_{j,k} \tag{1}$$

subject to

$$\sum_{a \in \delta^+(u)} f_{a,k} \; = \; \sum_{a \in \delta^-(u)} f_{a,k} \qquad \forall \, u \in \mathcal{N}, \; k \in \mathcal{K}, \tag{2}$$

$$\sum_{a \in \mathcal{A}} c_a \, f_{a,k} \; \leq \; T \qquad \forall \, k \in \mathcal{K}, \tag{3}$$

$$\sum_{a \in A_j} f_{a,k} \; \geq \; x_{j,k} \qquad \forall \, j \in \mathcal{J} \setminus \mathcal{J}_C, \; k \in \mathcal{K}, \tag{4}$$

$$x_{j,k} \; \leq \; y_j \qquad \forall \, j \in \mathcal{J} \setminus \mathcal{J}_C, \; k \in \mathcal{K}, \tag{5}$$

$$y_j \; \leq \; \sum_{k \in \mathcal{K}} x_{j,k} \qquad \forall \, j \in \mathcal{J} \setminus \mathcal{J}_C, \tag{6}$$

$$\sum_{k \in \mathcal{K}} x_{j,k} \; \leq \; 1 \qquad \forall \, j \in \mathcal{J} \setminus \mathcal{J}_C, \tag{7}$$

$$y_j \; = \; 1 \qquad \forall \, j \in \mathcal{J}_M, \tag{8}$$

$$\sum_{a \in \mathcal{A}} f_{a,k} \; \leq \; \mathcal{M} \sum_{a \in \delta^+(S)} f_{a,k} \qquad \forall \, k \in \mathcal{K}, \tag{9}$$

$$f_{a,k} \; \in \; \mathbb{Z}_{\geq 0} \qquad \forall \, a \in \mathcal{A}, \; k \in \mathcal{K}, \tag{10}$$

$$x_{j,k} \; \in \; \{0, 1\} \qquad \forall \, j \in \mathcal{J} \setminus \mathcal{J}_C, \; k \in \mathcal{K}, \tag{11}$$

$$y_j \; \in \; \{0, 1\} \qquad \forall \, j \in \mathcal{J} \setminus \mathcal{J}_C. \tag{12}$$

# 4 Model Description

The objective function (1) seeks to maximise the total length of cleaned streets minus a penalty for water waste. The first term sums the length of every street that is cleaned; the second term accounts for the excess water used when a vehicle's capacity exceeds the street's requirement, weighted by the penalty factor $W$. When $W = 0$ the problem reduces to pure coverage maximisation.

Constraint (2) enforces *flow conservation* at every junction for each vehicle: the total flow leaving a junction equals the total flow entering it. This ensures that each vehicle's route forms a closed walk, so it can return to its starting point without discontinuities.

Constraint (3) limits each vehicle's total traversal time to the shared time budget $T$, guaranteeing that every vehicle returns to the depot on time.

Constraint (4) requires that if vehicle $k$ cleans street $j$, it must have traversed at least one arc of that street. A vehicle cannot clean a street it has not physically visited.

Constraints (5) and (6) together link the per-vehicle cleaning variables $x_{j,k}$ to the aggregate cleaned indicator $y_j$. Constraint (5) ensures that cleaning by any vehicle activates the indicator ($x_{j,k} = 1 \Rightarrow y_j = 1$). Constraint (6) prevents the indicator from being active unless at least one

vehicle actually cleans the street ($y_j = 1 \Rightarrow \sum_k x_{j,k} \geq 1$). Together they encode $y_j = 1 \Leftrightarrow \exists\, k : x_{j,k} = 1$.

Constraint (7) limits each street to be cleaned by at most one vehicle. Although the problem permits redundant cleaning, enforcing single-vehicle cleaning eliminates unnecessary water waste, since a street's length is scored only once regardless of how many vehicles clean it.

Constraint (8) ensures that every mandatory street is cleaned. A solution that leaves any mandatory street uncleaned is infeasible. If a mandatory street is provably unreachable within the time budget (detected via a Dijkstra-based pre-check), its hard constraint is relaxed to a soft one so the solver is not forced into infeasibility.

Constraint (9) is a Big-M formulation that forces each vehicle to include the depot in its route. If a vehicle uses any arc (left-hand side $> 0$), it must have at least one outgoing traversal from the depot. Combined with flow conservation (2), this ensures the closed walk passes through $S$. When the depot has zero outgoing flow, the right-hand side collapses to 0, prohibiting all arc usage — i.e. an unused vehicle simply stays at the depot.

Constraints (10)–(12) state the variable domains: flow variables are non-negative integers, and cleaning and cleaned-indicator variables are binary.

This model is a Mixed-Integer Linear Programming problem representing a multi-vehicle arc routing problem with heterogeneous fleet, capacity matching, time windows, and a bi-criteria objective balancing street coverage against water efficiency, in the context of climate-aware urban street cleaning.

### 4.1 Subtour Elimination

Flow conservation alone does not prevent *subtours* — disconnected cycles that never visit the depot. These are eliminated via an iterative cutting-plane procedure. After each solve, the weakly-connected components of each vehicle's arc-flow multigraph are computed. For every component $\mathcal{C}$ that does not contain the depot, the following constraint is added:

$$\sum_{\substack{a \in \mathcal{A}: \\ \text{tail}(a) \in \mathcal{C},\, \text{head}(a) \in \mathcal{C}}} f_{a,k} \;\leq\; \left(1 + \sum_{\substack{a \in \mathcal{A}: \\ \text{tail}(a) \in \mathcal{C},\, \text{head}(a) \in \mathcal{C}}} U_a\right) \cdot \sum_{\substack{a \in \mathcal{A}: \\ \text{tail}(a) \notin \mathcal{C},\, \text{head}(a) \in \mathcal{C}}} f_{a,k} \tag{13}$$

This constraint states that the total flow on arcs internal to $\mathcal{C}$ can be positive only if there is positive flow on arcs entering $\mathcal{C}$ from outside. If no arcs enter the component, all internal flow is forced to zero. The model is then re-solved with the added cuts (warm-started from the previous solution) and the process repeats until no subtours remain or the time budget is exhausted.

### 4.2 Post-Processing: Route Reconstruction

After solving, each vehicle's arc-flow solution is converted into an explicit junction sequence via Hierholzer's algorithm for Eulerian circuits. If any connected component of a vehicle's multigraph is disconnected from the depot, a shortest-path bridge (depot $\rightarrow$ component $\rightarrow$ depot) is inserted when the remaining time budget permits; otherwise the component's arcs are dropped.